

Goldbach's Conjecture

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Abstract - We discuss the Goldbach Conjecture, a famously unsolved problem in number theory. One of its iterations states that every positive odd number greater than 7 can be written as the sum of three odd primes. In particular, we consider Christian Goldbach and Leonard Euler's correspondence and their early discussions of the Goldbach Conjecture. We also state derivations of the problem, including the ternary, strong, and weak versions of the Goldbach Conjecture. We also review the ideas behind the efforts to prove the conjecture, specifically some brute force attempts, Ivan Matveevich Vinogradov's theorem, and Harald Andrés Helfgott's recent proof on the weak Goldbach Conjecture. Finally, we examine the Goldbach Conjecture's significance and role in number theory and mathematics as a whole by relating it to other famously unsolved problems, such as the Riemann Hypothesis. We also explore how the conjecture can be used in other fields of mathematics, such as in cryptography and in group theory.

Keywords: cryptography; Goldbach Conjecture; group theory; number theory; prime numbers

1 INTRODUCTION

Prime numbers, positive numbers which are divisible by only 1 and themselves, have long captivated mathematicians. They have many practical applications, such as in cryptography, and appear in many famous theorems. One such theorem, the fundamental theorem of arithmetic, states that every positive integer larger than 1 is a unique product of prime numbers. What about addition? Is it possible to construct a summation of a limited number of primes to represent every number larger than 2? In 1742, Christian Goldbach wrote a letter to Leonhard Euler in which he proposes the following conjecture: every integer greater than 5 can be written as the sum of three primes. Euler responded with a similar conjecture: every even number greater than 2 can be written as the sum of two primes. These conjectures, known as the ternary Goldbach Conjecture and the strong Goldbach Conjecture respectively, have remained unproven since then. In this paper, we aim to discuss the history of this famous conjecture including its author, variations, and initial proposition. Additionally, we showcase various attempts to prove the conjecture and reasons why this conjecture is so difficult to prove. Finally, we will explore the consequences and the conjecture's uses in number theory and other fields.

2 HISTORY OF GOLDBACH'S CONJECTURE

Christian Goldbach was a Prussian mathematician in the 1700s. Born in 1690, Goldbach studied law and took an interest in mathematics. His main research was in number theory and he had a close friendship with Leonhard Euler, with whom he exchanged many letters. Their correspondence contains many of Goldbach's most important contributions to number theory.

One particular letter, sent on June 7th, 1742 by Goldbach to Euler, proposes the following:

Goldbach's First Conjecture: Every integer greater than 2 can be written as the sum of three primes.

On June 30th of the same year, Euler responded with a related conjecture,

Every positive even integer can be written as the sum of two primes.

In this same letter, Euler stated, “That every even number is a sum of two primes, I consider an entirely certain theorem in spite of that I am not able to demonstrate it.” [1].

Note that both Goldbach and Euler used the now-abandoned convention of considering 1 to be a prime number. We restate the conjectures using updated conventions as follows:

Ternary Goldbach Conjecture: Every integer greater than 5 can be written as the sum of three primes.

The above conjecture, often called the Ternary Goldbach Conjecture can be easily tested for small numbers. For example,

$$6 = 2 + 2 + 2,$$

$$7 = 2 + 2 + 3,$$

$$8 = 2 + 3 + 3,$$

and so on.

Strong Goldbach Conjecture: Every positive even integer greater than 2 can be written as the sum of two primes.

This conjecture is often called the Strong Goldbach Conjecture. We illustrate the conjecture on the first few even integers.

$$4 = 2 + 2,$$

$$6 = 3 + 3,$$

$$8 = 5 + 3.$$

Unlike the fundamental theorem of arithmetic, this conjecture does not guarantee uniqueness. For example,

$$10 = \begin{cases} 5 + 5 \\ 3 + 7 \end{cases} \qquad 26 = \begin{cases} 23 + 3 \\ 19 + 7 \\ 13 + 13 \end{cases}$$

We mention also a third, “weaker” version of the conjecture:

Weak Goldbach Conjecture: Every positive odd number greater than 7 can be written as the sum of three odd primes.

We demonstrate the conjecture. Note again that this conjecture does not guarantee uniqueness.

$$9 = 3 + 3 + 3,$$

$$11 = 3 + 3 + 5,$$

$$13 = 5 + 5 + 3,$$

$$15 = 5 + 5 + 5 = 5 + 7 + 3.$$

Interestingly, René Descartes made a similar conjecture many years prior to Goldbach’s and Euler’s correspondence. He said,

Descartes's Conjecture: Every even number can be expressed as the sum of at most three primes.

Paul Erdős said of this, “Descartes actually discovered this before Goldbach, but it is better that the conjecture was named for Goldbach because, mathematically speaking, Descartes was infinitely rich and Goldbach was very poor.” [2]

Regardless of who the conjecture should be named after, neither mathematician could prove the conjecture, and it remains an open problem.

3 PROOF ATTEMPTS

Since this conjecture was originally made, many have attempted to prove it. The strong Goldbach Conjecture is the most difficult to prove. The most progress has been made on the weak Goldbach Conjecture.

In 1938, Nils Pipping took on the task of verifying the strong Goldbach Conjecture up until $n = 100,000$. More recently, Oliveira e Silva [3] ran a computer search for up until $n = 4 \cdot 10^{18}$ and double-checked the strong Goldbach Conjecture up until $n = 4 \cdot 10^{17}$. Tomás Oliveira e Silva accomplished this by searching for Goldbach partitions, defined as follows,

Goldbach Partitions: Let n be an even number larger than two, and let $n = p + q$, with p and q prime numbers such that $p \leq q$, be a Goldbach partition of n .

His strategy was to search for the minimal Goldbach partition, which he defined to be the one that uses the smallest possible prime number p . For example, the minimal Goldbach partition of 10 would be $p = 3, q = 7$ instead of $p = 5, q = 5$.

Despite this vast search, there is no definitive proof of the conjectures. One of the first big strides in attempting to verify the conjecture came from Ivan Matveevich Vinogradov, with his famous three prime theorem. Before stating the theorem, we define the following two functions:

van Mangoldt function [4]:

We define the following function

$$\Lambda(n) = \begin{cases} \log p & \text{if } n = p^k \text{ for a prime number } p, \\ 0 & \text{otherwise} \end{cases}$$

and call it the van Mangoldt function.

$r(n)$ [4]:

We define the function $r(n)$ is the number of ways to express n as the sum of three numbers. These numbers are either prime or a power of a prime with a weight of $\log(p_1) \log(p_2) \log(p_3)$ attached to that value. Then,

$$r(n) = \sum_{k_1 + k_2 + k_3 = n} \Lambda(k_1) \Lambda(k_2) \Lambda(k_3)$$

Note that if $r(n) \neq 0$, we do not have a guarantee that there exists a representation of an integer as a summation of three prime numbers. We only have that a representation of three prime powers exists. However, since the prime powers contribute less than the prime numbers to the summation, bounding this function is enough [4].

Vinogradov's proof is beyond the scope of this paper, but we briefly outline his techniques. Vinogradov used the Hardy-

Littlewood circle method, a powerful method in analytic number theory. It uses Fourier analysis to detect patterns in prime numbers. The idea is to express a counting problem using exponential sums and dividing the unit circles into major and minor arcs. These arcs capture the main structures of primes.

Using these techniques, Vinogradov was able to determine an asymptotic estimate for $r(n)$.

Vinogradov's Theorem [5]:

For any fixed $A > 0$,

$$r(N) = \frac{1}{2} \mathfrak{S}(N) N^2 + O\left(\frac{N^2}{\log^A(N)}\right),$$

where

$$\mathfrak{S}(N) = \prod_{p|N} \left(1 - \frac{1}{(p-1)^2}\right) \prod_{p \nmid N} \left(1 + \frac{1}{(p-1)^3}\right)$$

Note that if N is even, $\mathfrak{S}(N) = 0$ and the above bound is meaningless. We consider the case when N is odd.

Lemma 3.1 [5]:

When N is odd, we have

$$\prod_{p \neq 2} \left(1 - \frac{1}{(p-1)^2}\right) = 0.6602 \dots,$$

and

$$\prod_p \left(1 + \frac{1}{(p-1)^3}\right) = 2.301 \dots,$$

thus, $1.320 \dots < \mathfrak{S}(N) < 2.301 \dots$. Then, we have that the N^2 term will eventually dominate the second term. As a result, $r(N)$ has a lower bound.

This lower bound allows us to form the following corollary.

Corollary 3.1: Every sufficiently large odd integer is the sum of three primes.

Proof: By Lemma 3.1, $\mathfrak{S}(N)$ is bounded for all odd N , so for sufficiently large N , N^2 is much larger than $\mathfrak{S}(N)$. Then, $r(n)$ is bounded below for sufficiently large N . Since $r(n)$ is the number of representations of n as the sum of three prime numbers, we have that for sufficiently large values of N , N has a representation as the sum of three prime numbers [4]. $\square$

In Vinogradov's original proof, he did not give a definition for "sufficiently large". However, more recent estimates show that 10^{1346} is sufficiently large for this theorem to hold. Additionally, Yannick Saouter implemented an algorithm to check the weak Goldbach Conjecture for numbers up until 10^{20} [6]. Then, there remain finitely many numbers to check to prove the weak version of Goldbach's Conjecture [7].

More recently, Harald Andr s Helfgott submitted a paper in which he claims to have lowered the "sufficiently large" requirement to 10^{27} and checked the remaining numbers until then, thus proving the weak Goldbach Conjecture [8].

His 240-page proof is again beyond the scope of this paper. His approach is similar to Vinogradov's, using Fourier analysis and exponential sums. However, he was able to set major arcs to be few and narrow, hence reducing the "sufficiently large" condition significantly. This proof was accepted for publication at Annals of Mathematics Studies in 2015 and Helfgott has been editing and compiling his proof into a book since. As of July 7th, 2024, he is still working on the book [9].

What remains to be proven are the stronger versions of Goldbach's Conjecture, including the ternary Goldbach Conjecture and the strong Goldbach Conjecture. The difficulty in proving these conjectures might lie in the challenging nature of prime numbers. Their distribution across the natural numbers does not follow a simple pattern and we are not able to predict where prime numbers lie directly. It could also be the case that the conjecture is simple to prove, but we have not yet approached the problem correctly. Some believe that the conjecture might be unprovable.

Donald Knuth said, "Goldbach's Conjecture is just, sort of, true because it can't be false. There are so many ways to represent an even number as the sum of two odd numbers, that as the numbers grow the number of representations grows bigger and bigger. Take a $10^{10^{10}}$ -digit even number, and imagine how many ways there are to write that as the sum of two odd integers. For an n -bit odd number, the chances are proportional to $1/n$ that it's prime. How are all of those pairs of odd numbers going to be nonprime? It just can't happen. But it doesn't follow that you'll find a proof, because the definition of primality is multiplicative, while Goldbach's Conjecture pertains to an additive property. So it might very well be that the conjecture happens to be true, but there is no rigorous way to prove it." [10]

4 SIGNIFICANCE IN NUMBER THEORY

The Goldbach Conjecture may seem to be a simple statement that just happens to have stumped mathematicians for ages. However, the conjecture is connected to other notoriously hard problems such as the Riemann Hypothesis and the Twin Prime Conjecture.

Riemann Hypothesis:

Consider the Riemann zeta function defined as

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}$$

with $\text{Re}(s) > 1$, then all nontrivial zeros lie on the critical line, such that $s = \frac{1}{2} + it$, $t \in \mathbb{R}$.

Twin Prime Conjecture: There are infinitely many twin primes.

Twin Primes: A twin prime is a pair of prime numbers $(p, p + 2)$.

All of these unproven statements theorize on the distribution of prime numbers. While each one focuses on different aspects of prime number distribution, proving any of them would give insights into proving the others. In particular, proving Goldbach's Conjecture would strengthen conjectures regarding gaps between prime numbers.

It would also refine our understanding of cryptography, an entire branch of mathematics and computer science which is heavily reliant on prime numbers. For instance, we may develop new cryptosystems using Goldbach's Conjecture.

M.K. Viswanath and M. Ranjith Kumar developed an authentication protocol using Goldbach's Conjecture, as well as Vinogradov's work, and decimal expansion of irrational numbers. [11]

Feasible Numbers [11]: Feasible numbers satisfy one of the following two conditions:

- (1) Suppose N is a large even number, then N satisfies the decomposition $N = P + r_1 \cdot s_1$, where r_1 and s_1 are distinct primes and P is the largest prime satisfying this relation.
- (2) If M is a large odd integer, then M satisfies the decomposition $M = Q + r_2 + s_2$, where r_2 and s_2 are appropriate distinct primes and Q is the largest prime satisfying this relation.

Chen's Theorem: A large even integer can be expressed as the sum of a prime and the product of at most two primes.

Note that P and Q are guaranteed to exist through a combination of Chen's theorem and Vinogradov's theorem.

Suppose Alice and Bob are exchanging messages. Bob may ask Alice to send him a large feasible even number, N_1 . Then N_1 has the decomposition $N_1 = P_1 + r_1 \cdot s_1$, where $r_1 < s_1$ and P_1 is the largest prime which satisfies this decomposition. Bob sends a large feasible number M_1 to Alice such that $M_1 = Q_1 + r_2 + s_2$ with $r_2 > s_2$, $s_1 = s_2$ and Q_1 is the largest prime which satisfies this decomposition. Both Alice and Bob can compute (P_1, r_1, s_2) and (Q_1, r_2, s_2) . They also choose an irrational number I with a decimal expansion up to more than one million decimal places.

Using these values, modular arithmetic, as well as rectangular matrices and their pseudo inverses, Viswanath and Ranjith Kumar created a secure and alternative cryptosystem to those used today.

We also discuss the use of Goldbach's Conjecture in developing a fast algorithm for identifying gray box groups. These groups behave similarly to the symmetric group S_n and alternating group A_n , but the inner structure is not completely visible.

This method identifies whether a given grey box group is isomorphic to S_n or A_n when $n \geq 20$. Sergey Bratus and Igor Pak extend the Goldbach Conjecture in two ways:

Extended Goldbach Conjecture [12]:

For every even $n \geq 20$, there exist primes $p_1 > p_2 > 3$ such that $p_1 + p_2 = n$. Moreover, the number $\pi_2(n)$ of such pairs is

$$\pi_2(n) > \frac{1}{3} \frac{n}{\log^2 n}.$$

Weighted Goldbach Conjecture [12]:

There exists a positive constant $c > 0$ such that for all even $n \geq 20$ we have

$$P_n > \frac{1}{4} \frac{\log \log n}{n \log n},$$

where $P(n)$ is equal to the proportion of Goldbach elements in S_n .

Goldbach Elements [12]: An element $y \in S_n$ is a Goldbach element if its order, denoted by $r(y)$, satisfies

$$r(y) = p_1 p_2,$$

where p_1, p_2 are distinct primes such that $p_1 + p_2 = n$.

We discuss the algorithm for a symmetric group S_{2k} .

Algorithm for S_{2k} [12]:

Let $n = 2k$.

- (1) Locate a Goldbach element in G . We find this element $y \in G$ with $r(y) = p_1 p_2$, such that p_1, p_2 are distinct primes and $p_1 + p_2 = n$. Note that by the extended Goldbach Conjecture, there will be one such pair of primes.

Then, the element y consists of two cycles of lengths p_1 and p_2 .

$$y_1 = (i_1, i_2, \dots, i_{p_1}),$$

$$y_2 = (j_1, j_2, \dots, j_{p_2})$$

Then renumber the points so that

$$\{i_1, i_2, \dots, i_{p_1}\} = 1, \dots, p_1 \text{ and } \{j_1, j_2, \dots, j_{p_2}\} = \{p_1 + 1, \dots, n\}.$$

- (2) Locate a transposition in G . We search for $x \in G$ such that $r(x) = 2q_1 q_2$ where $q_1 + q_2 = n - 2$ and q_1, q_2 are distinct odd primes. Note again that there will be one such pair of primes due to the Goldbach Conjecture. Then, x consists of 2 odd prime cycles and a transposition. Namely,

$$a = x^{q_1 q_2}$$

is a transposition.

- (3) Check if the transposition touches both y_1 and y_2 . If not, produce and try random transpositions until we have this property. We search for a transposition that interchanges a point from y_1 and a point from y_2 . Then, we check that

$$ay_1 \neq y_1 a \text{ and } ay_2 = y_2 a.$$

- (4) Compute $b = y_1 a y_2$. If $a = (l_1, l_2)$ does not commute with either cycle, then one of l_1 and l_2 lies in y_1 and the other in y_2 . Assume, without loss of generality, that

$$\{l_1, l_2\} = \{p_1, p_1 + 2\}.$$

Then b is a cycle of length $p_1 + p_2 = n$ and that there is a homomorphism $\phi : G \rightarrow S_n$ that carries b into $(1, 2, \dots, n)$ and a into $(p_1, p_1 + 1)$.

- (5) Conjugate a by b^{p_1-1} to obtain a' .

$$\phi(b) = (1, 2, \dots, n), \phi(a) = (p_1, p_1 + 1), \phi(a') = (1, 2),$$

and as $\{a, b\}$ generate S_n , we have the isomorphism $\phi : G \rightarrow S_n$ under which we can compute the image of any element $g \in G$.

The algorithm for S_{2k+1} is similar, but we must find two transpositions instead of one. The constructed isomorphism is critical in identifying whether the grey box group is isomorphic to S_n .

By assuming these stronger versions of the Goldbach Conjecture, Bratus and Pak were able to find the worst-case estimate of their algorithm to be $O([\rho + \nu + \mu]n \log^2 n)$ or $O([\rho + \nu + \mu]n \log n / \log \log n)$ assuming the extended Goldbach Conjecture and weighted Goldbach Conjecture respectively. [12]

Goldbach's Conjecture proves useful not only in additive number theory but also in other fields such as cryptography and group theory. Additionally, being able to prove this conjecture may bring about new methods that would allow us to prove other conjectures in number theory.

5 CONCLUSION

Goldbach's Conjecture remains one of the most elusive problems in number theory. Since its initial proposal in 1742, mathematicians have attempted to prove the conjecture, coming as close as proving the weak version of Goldbach's Conjecture in 2013. We have shown that it is possible to construct a summation of three primes to represent odd numbers larger than 7. We have yet to show that every positive even number greater than 2 can be written as the sum of two primes. The conjecture's close connection to prime number distribution makes it difficult to prove. Even without having been proven, conjecture has been used to deepen understanding of cryptography and develop new algorithms for object identification in group theory. Whether it is rigorously proved or not, the investigation into this conjecture continues to deepen our understanding of prime numbers and their role in mathematics.

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